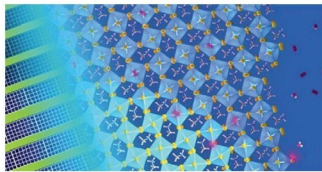


of Things (IoT) devices used in medical devices. So someone with commercially-available equipment could send commands to the pacemaker, changing its settings and software. The 'patched' version will prevent this by allowing only authorised hardware and software tools to send commands to the device.

Defects in next-generation solar cells can be healed with light

Researchers have shown that defects in the molecular structure of perovskites—a material which could revolutionise the solar cell industry—can be healed by exposing it to light and just the right amount of humidity. This finding by the team of researchers from Cambridge, MIT, Oxford, Bath and Delft could further accelerate the development of cheap, high-performance perovskite-based solar cells that rival the efficiency of silicon.



The concoction of light with water and oxygen molecules leads to substantial defect-healing in metal halide perovskite semiconductors (Image courtesy: Dr Matthew T. Klug)

Tiny defects in the crystalline structure of perovskites, called traps, can cause electrons to get 'stuck' before their energy can be harnessed. In their study, the researchers made a perovskite-based device, printed using techniques compatible with scalable roll-to-roll processes. Before the device was completed, they exposed it to light, oxygen and humidity. Perovskites often start to degrade when exposed to humidity, but the team found that when humidity levels were between 40 and 50 per cent, and the exposure was limited to 30 minutes, degradation did not occur. Once the exposure was complete, the remaining layers were deposited to finish the device.

When the light was applied, electrons bound with oxygen, forming a superoxide that could very effectively bind to electron traps and prevent these traps from hindering electrons. In the accompanying presence of water, the perovskite surface also got converted into a protective shell. The shell coating removed traps from the surfaces but also locked in the superoxide, meaning that the performance improvements in the perovskites were now long-lived.

A new approach to ultrafast light pulses

Two-dimensional materials called molecular aggregates are very effective light emitters that work on a different principle than typical organic light-emitting diodes (OLEDs) or quantum dots. But their potential as components for new kinds of optoelectronic devices has been limited by their relatively slow response time. Now, researchers at MIT, the University of California at Berkeley, and Northeastern University have found a way to overcome that limitation, potentially opening up a variety of applications for these materials.

The team found that the key to enhancing the response time of these 2D molecular aggregates (2DMA) is to couple that material with a thin layer of a metal such as silver. The interaction between the 2DMA and the metal that is just a few nanometres away boosts the speed of the material's light pulses more than tenfold.

In order for 2DMA materials to be useful in devices such as photonic chips—which are like semiconductor chips but carry out their operations using light instead of electrons—the challenge is to be able to switch them on and off quickly.

With the metal substrate nearby, the response time for the light emission dropped from 60 picoseconds (trillionths of a second) to just 2 picoseconds. If used for signal processing, such as sending data by light rather than radio waves, this advance could lead to a data transmission rate of about 40 gigahertz, which is eight times faster than such devices can currently deliver.

World's cheapest and portable ventilator unveiled

India-based A-Set Robotics claims to have created the smallest and cheapest indigenous portable ventilator. Launched at AIIMS, New Delhi, this is the world's first ventilator to have artificial intelligence algorithm. It is so small that it can be easily carried in the pocket.

A-Set head and young scientist Diwakar Vaish shared that the ventilator is at least 50 times cheaper than conventional ventilators. Conventional ventilators cost ₹ 500,000 to 1.5 million each and sometimes even more. The A-Set ventilator, on the other hand, costs just ₹ 15,000-₹ 20,000. Also, it has a footprint that is 100 times smaller than conventional ventilators. It runs without use of oxygen cylinders, so running cost is almost nil. What's more, its machine learning technology learns how the patient breathes. Thus, in case of no biological feedback, the ventilator can send in air as per previously learned values.

The ventilator can be controlled by just a simple mobile application. Safety features include text messaging to the attendant in case of emergency, on-board light, diagnosis information relay to the manufacturers so that they can troubleshoot the problem and provide telephonic help, on-board buzzer to alert people around, and dual computers on-board so that in case one fails, the other one is always in standby to take over.